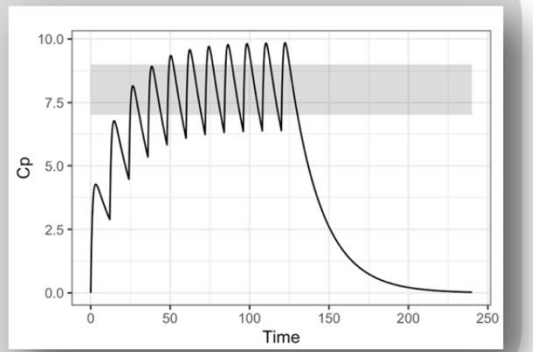
Virtual Patient Simulation

Curtis K. Johnston & Shen Cheng

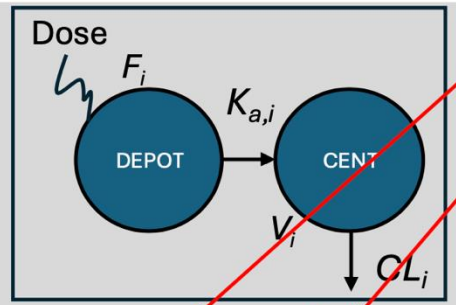
11/20/2026

Stochastic Simulation

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	...
65	31	F	...
77	25	M	...
102	36	M	...
...	...	...	...



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

...

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}}\right)$$

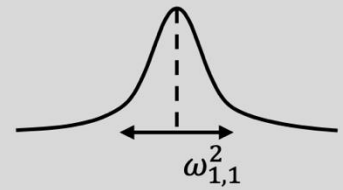
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

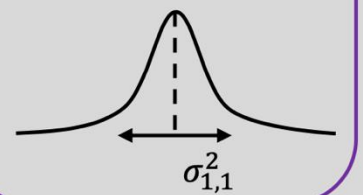
...

Population parameters (`.ext`)

1. Fixed-effect
   $\boldsymbol{\theta_1}, \boldsymbol{\theta_2}$ ...
2. Random-effect
   - 2.1 Ω matrix

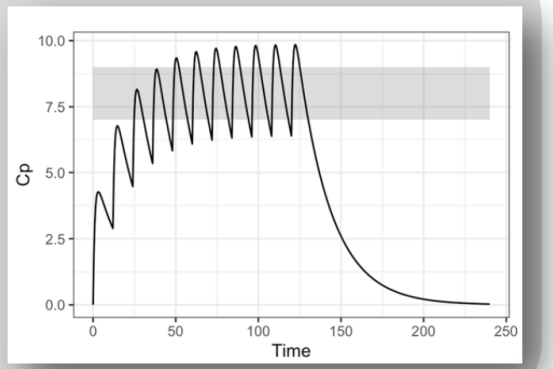


   - 2.2 Σ matrix

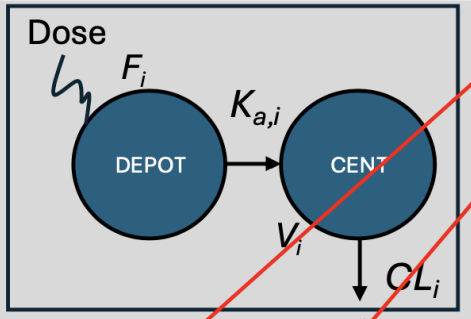


Covariate (Virtual Patient) Simulations

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
45	12	M	…
57	15	F	…
33	6	F	…
…	…	…	…



$$CL_i = \theta_1 \times \left(\frac{WT_i}{70kg}\right)^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}}\right)$$

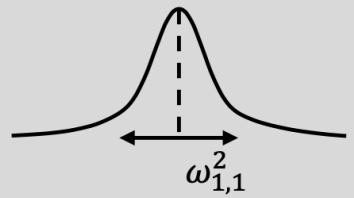
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

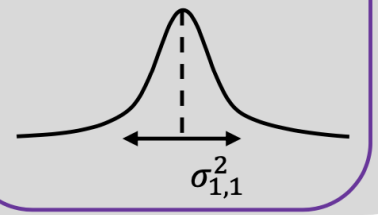
…

Population parameters (.ext)

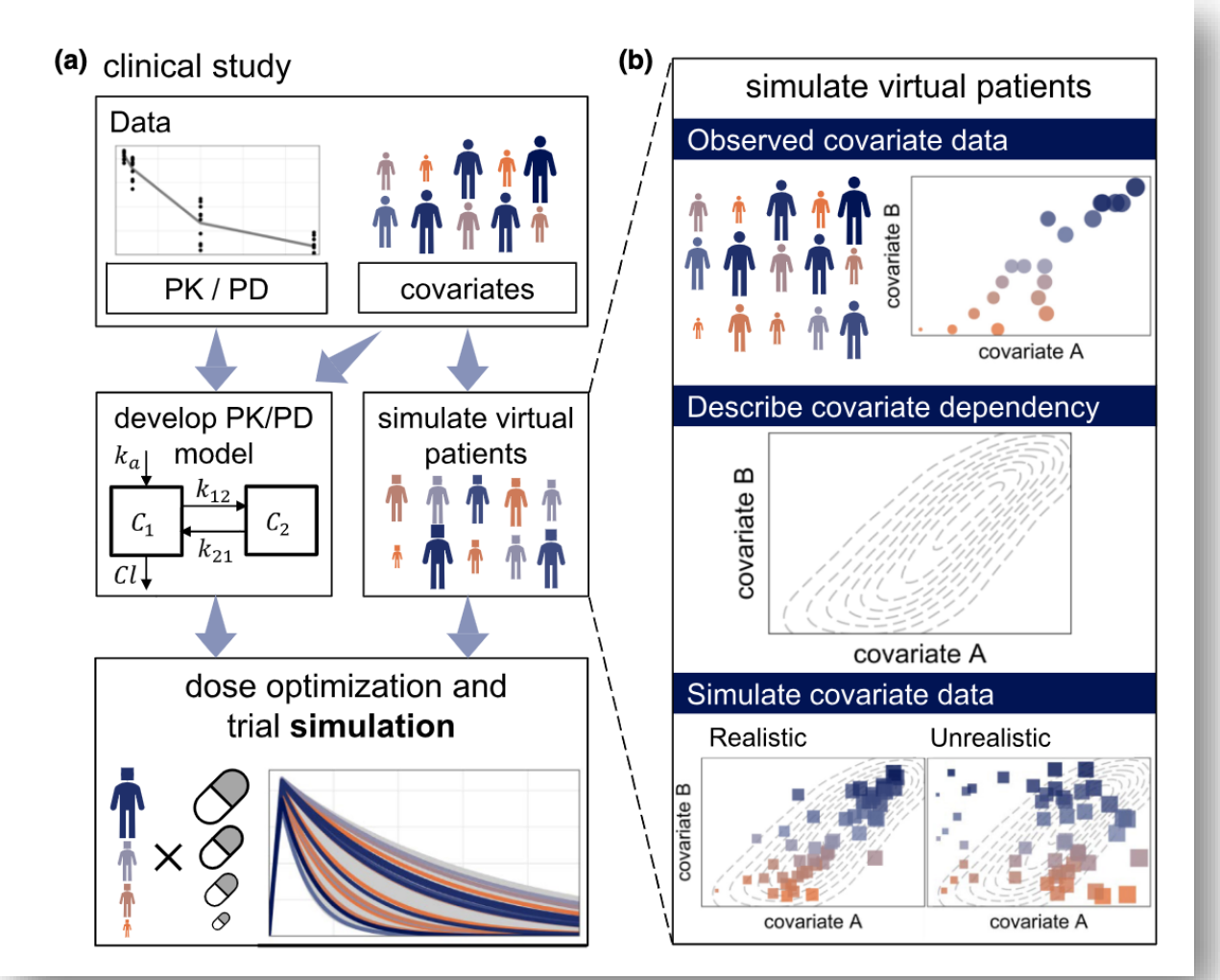
1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix



“Realistic” Virtual Population



Zwep LB et al., Clin Pharmacol Ther. 2023

Difficulties in simulating **realistic** virtual population:

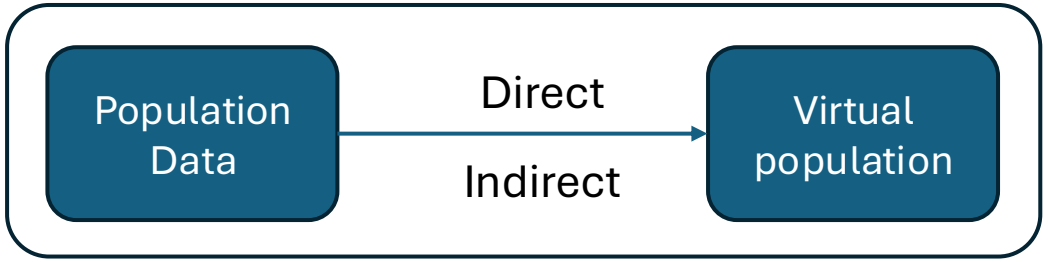
- Different covariates exhibit different distributions (i.e., marginal distributions)
- Intricate dependency structure (i.e., correlation)

Unrealistic virtual patient example:
A patient of 95 years old, with a high body weight and a very good kidney.

Failing to consider such correlations inflate the variability of covariates, in other words, unrealistic virtual patients.

Methods of Virtual Patient Simulations

- Data (Resample)-based methods
  - Bootstrap
  - Conditional distributions
    - Also known as Multiple Imputation by Chained Equations (MICE)



- Distribution-based methods
  - Marginal distributions
  - Multivariate normal distributions (MVND)
  - Copula modeling



Methods of Virtual Patient Simulations

Bootstrap

	X_1	X_2
1	4.5	50
2	10.3	32
3	3.5	16
4	0.2	103

Resample rows

	X_1	X_2
3	3.5	16
2	10.3	32
1	4.5	50
2	10.3	32

Need actual data
Cannot extrapolate

Conditional distributions (MICE)

Smania & Jonsson, 2021

Add missing rows

	X_1	X_2
1	4.5	50
2	10.3	32
3	3.5	16
4	0.2	103
5		
6		
7		
8		

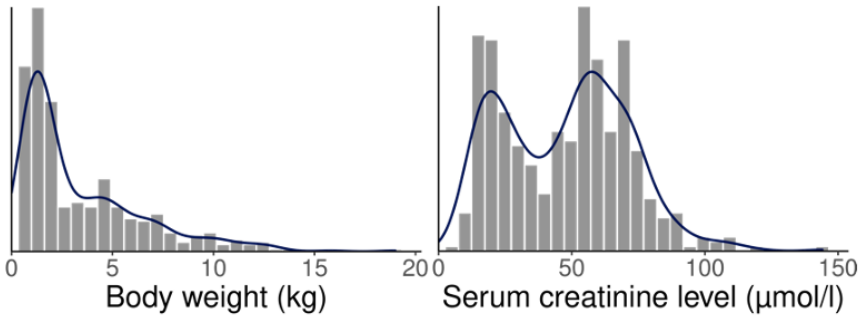
Impute missing rows

	X_1	X_2
1	4.5	50
2	10.3	32
3	3.5	16
4	0.2	103
5	3.9	70
6	16	35
7	0.7	59
8	12	9

Need actual data (indirectly)

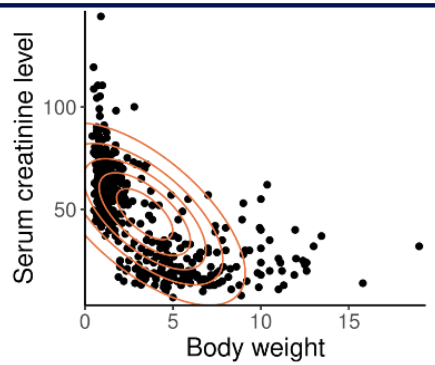
data-based

Marginal distributions



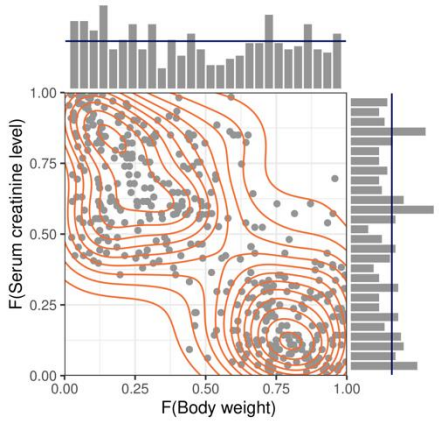
Ignore correlations

Multivariate normal distribution



Distributional assumptions

Copula modeling



distribution-based

Zwep LB et al., PAGE. 2022

Resample-Based Methods

- **Bootstrap**
- Conditional Distribution/Multiple Imputations

Bootstrap

- The distribution among covariates can be complicated.
- Knowing the underlying distribution among covariates may need some efforts.
- But we don’t need to know the data-generating distribution in order to create a population that follows the distribution
- Resampling allows us to generate a population that maintains the underlying distribution



Curtis Johnston. Population Generation via Resampling

How to resample?

ID	SEX	WT	AGE	ALB	EGFR
1	female	58.3	49	4.6	100.5
2	female	61.8	43	4.1	136.2
3	male	66.4	36	4.4	106.6
4	male	75.5	39	3.9	105.2
5	female	59.1	55	4.0	124.6
…					



ID	SEX	WT	AGE	ALB	EGFR
X1	female	59.1	55	4.0	124.6
X2	male	66.4	36	4.4	106.6
X3	female	59.1	55	4.0	124.6
X4	female	58.3	49	4.6	100.5
X5	male	75.5	39	3.9	105.2
…					

How to resample?

- In most instances resampling from your analysis population should suffice, but not always
  - If the population of interest is not well represented in your analysis dataset
    - Rules of thumb often suggest having at least 50 to 70 subjects per covariate category (such as genotype, disease state, or age group) (Dosne et al., 2016), but often hard to achieve in a population analysis.
    - Sometimes this is all you have and you just acknowledge the limitation
  - If the population of interest is not represented (at all) in your analysis dataset
    - Use publicly available research datasets (NHANES)

What is NHANES?

- National Health and Nutrition Examination Survey (NHANES)
  - Survey research program that collect longitudinal data on medical data and physiological measurements (lab data)
    - Surveys have been conducted annually since 1999 (first conducted in 1971)
  - Survey results are publicly available; either directly from the NCHS website or via convenient R packages.
    - https://wwwn.cdc.gov/nchs/nhanes/
    - nhanesA (R package): Introducing nhanesA

nhanesA R package example codes

nhaneA-example-codes.R

Source on Save | Run

```r

# Library
library(tidyverse)
library(nhanesA)

# Download demographic data (DEMO_D), and select/rename variables
dem <- nhanes('DEMO_D') %>%
    select(ID = SEQN, AGE = RIDAGEYR,
           SEX = RIAGENDR, RACE = RIDRETH1)

# Download body measure examination data (BMX_D), and select/rename variables
wt <- nhanes('BMX_D') %>%
    select(ID = SEQN, WT = BMXWT)

# Merge data
covar <- inner_join(dem, wt, by="ID")

# Remove NAs
covar <- filter(covar, !is.na(WT) & !is.na(AGE) & !is.na(SEX) & !is.na(RACE) )
anyNA(covar)

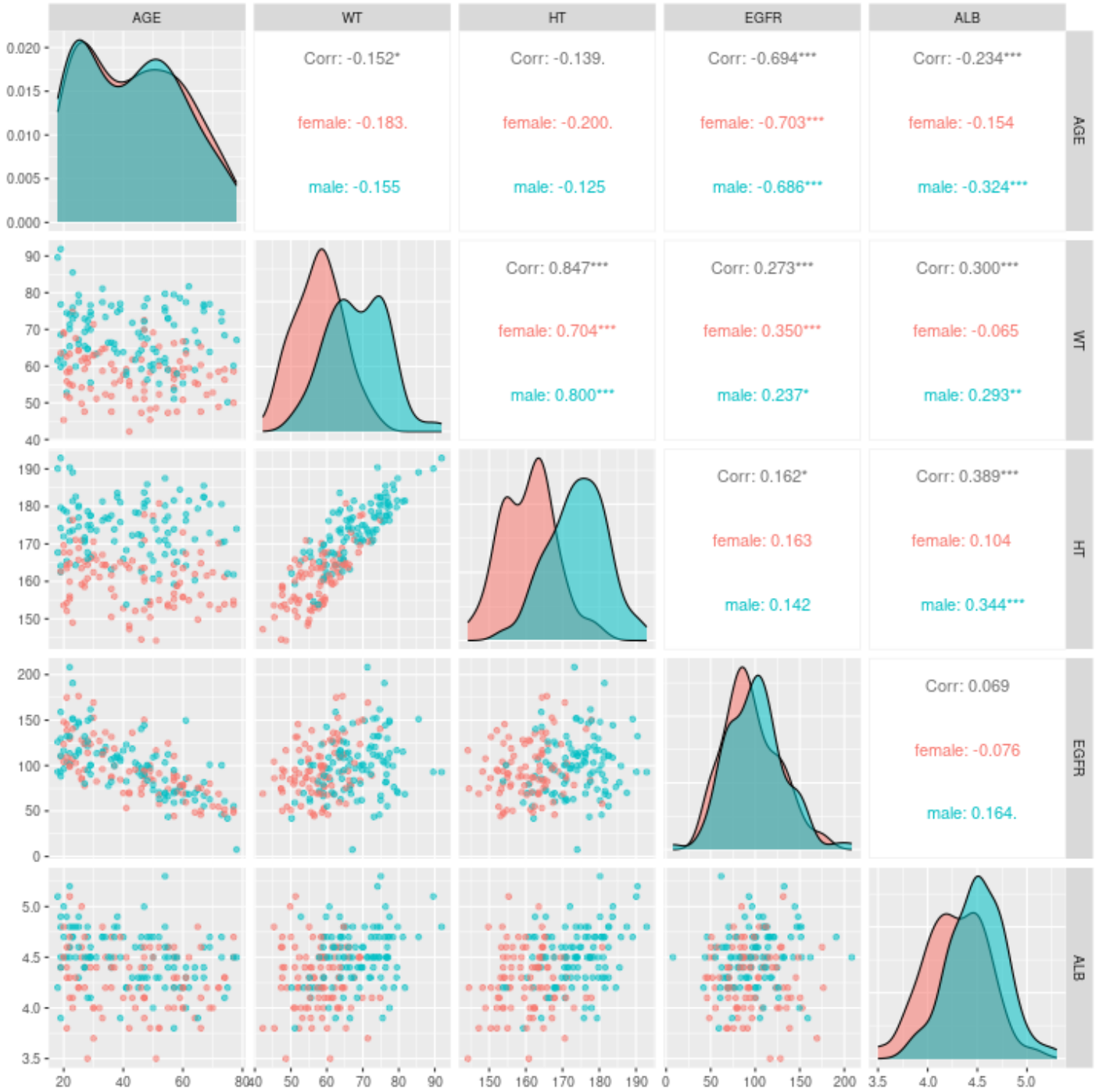
# Create virtual population (n=1000) via resampling with replacement
vp <- covar %>% sample_n(size = 1000,replace = T)

```

`./wk11/script/nhanesA-example-codes.R`

Distribution-Based Methods

- Marginal distributions
- Multivariate normal distributions (MVND)
- Copula modeling



Population Data

./wk11/data/pop.csv

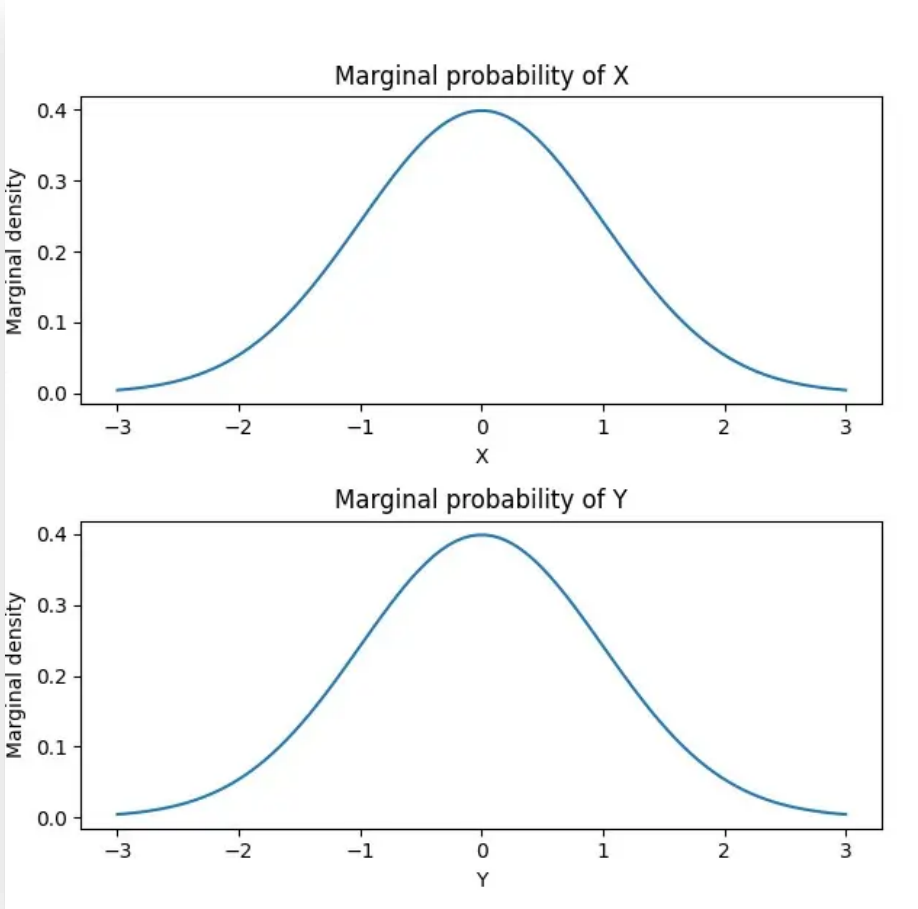
Seven covariates:

- Continuous:
  - AGE: age in years
  - WT: body weight in kg
  - HT: height in cm
  - EGFR: estimated glomerular filtration rate in mL/min
  - ALB: albumin concentration in g/dL
- Categorical:
  - SEX: gender (male vs female)

Experiments

- Separate the population data based on SEX (**male** vs **female**)
- Derive two distributions, one for male, one for female, using different method:
  - Marginal distributions
  - Multivariate normal distributions
  - Copula modeling
- Simulate from the derived distributions 100 times for simulation method evaluation.
- Final simulation from the derived distributions(n=1)
- Compare the simulated data with the population data, determine which method simulate the most "**realistic**" data (i.e., most similar to population data).

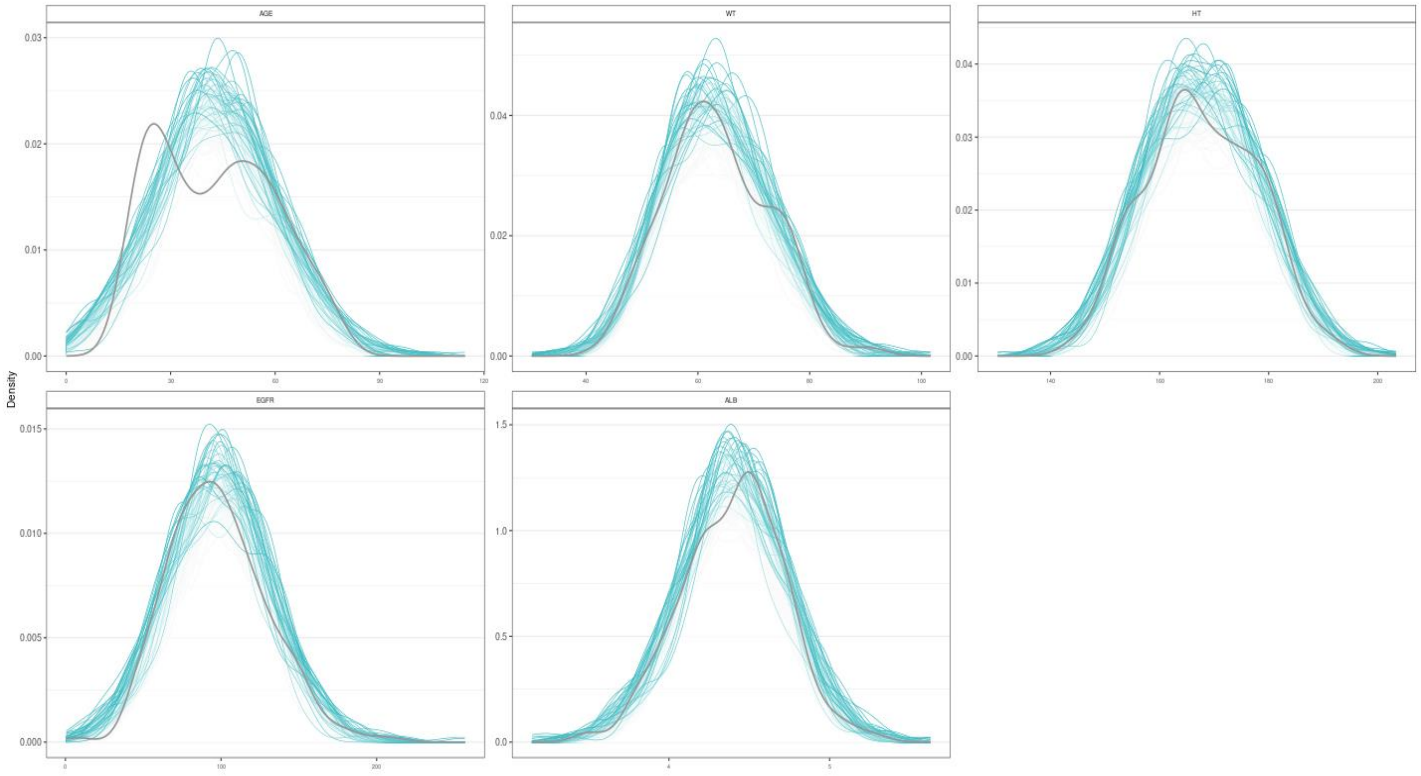
Marginal Distributions

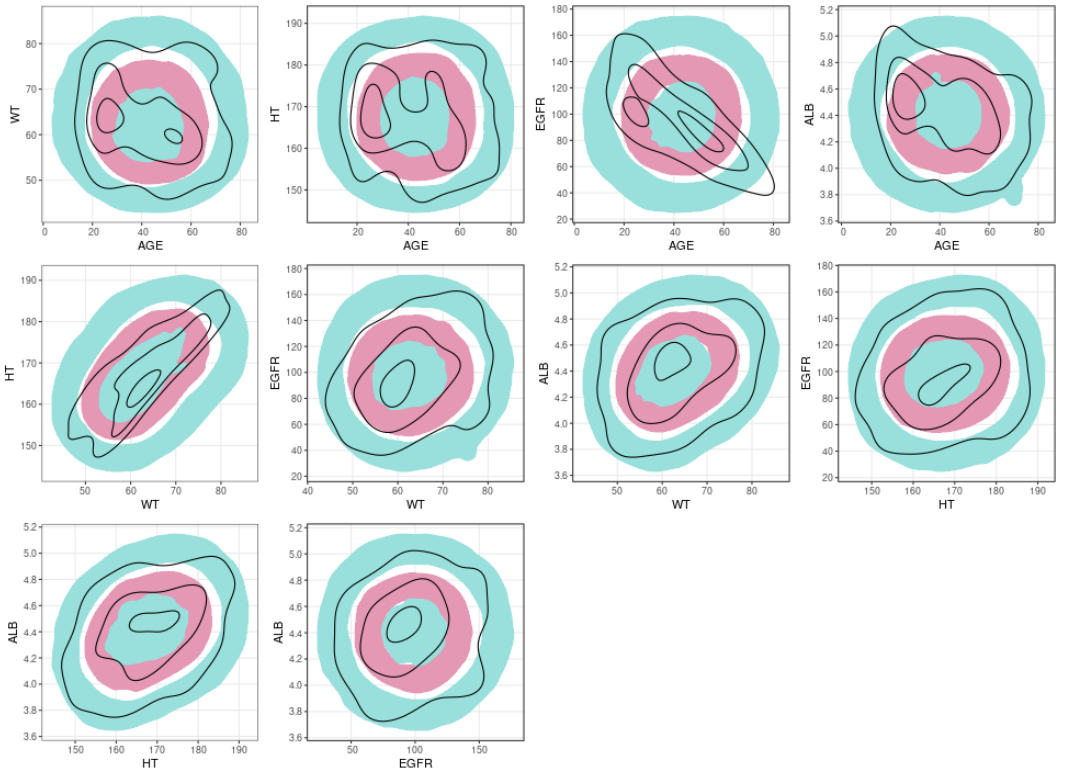


- Derive the means and standard deviations (SDs) of each variable in the population dataset.
- Sample from a (truncated) normal distribution (or other distributions) with the same mean and SD of each variable.

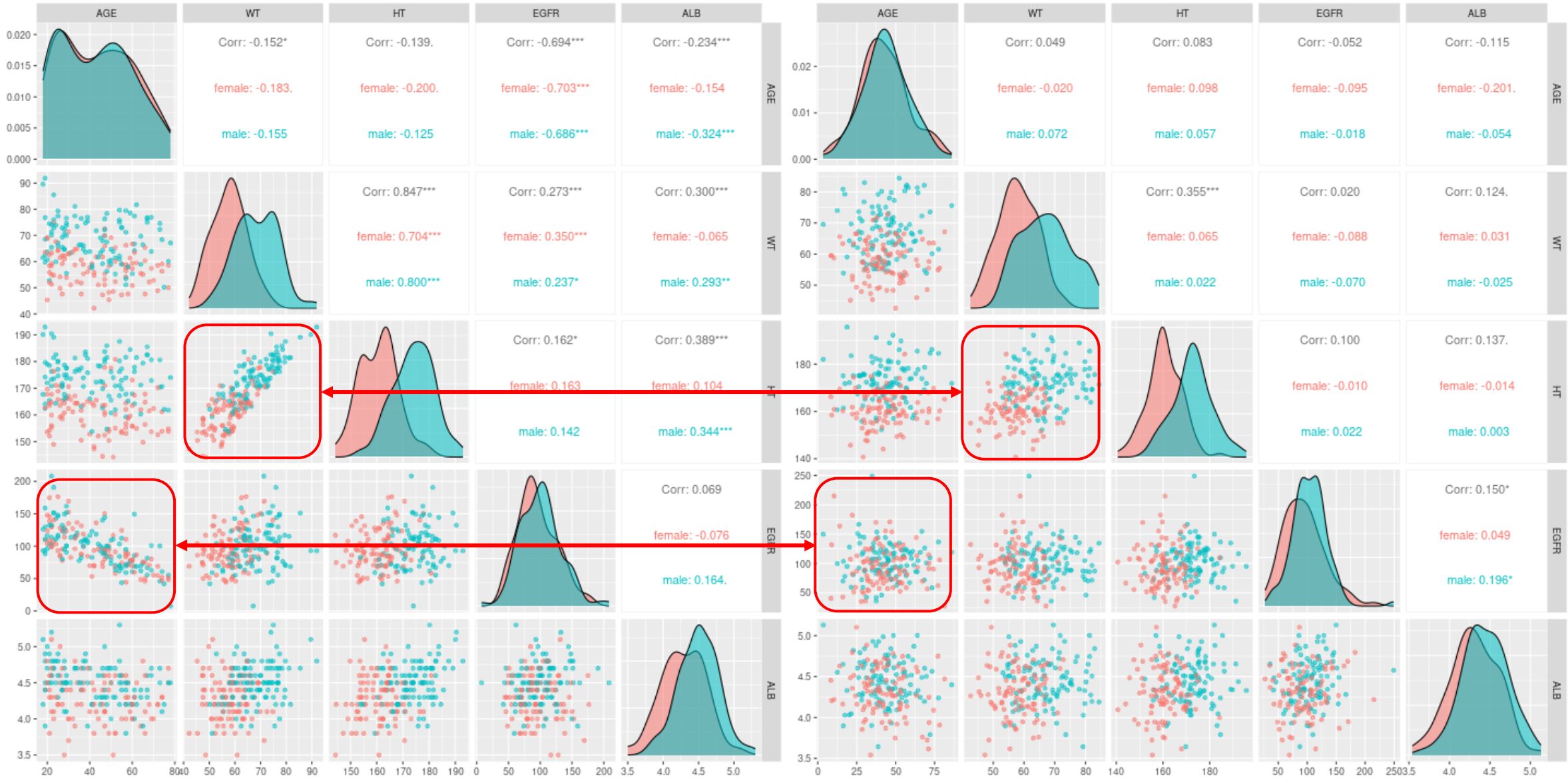
Meccanismo Complesso. Marginal probability. 12/11/2023

Marginal Distributions Simulation Method Evaluation (Simulation replicates: n=100)





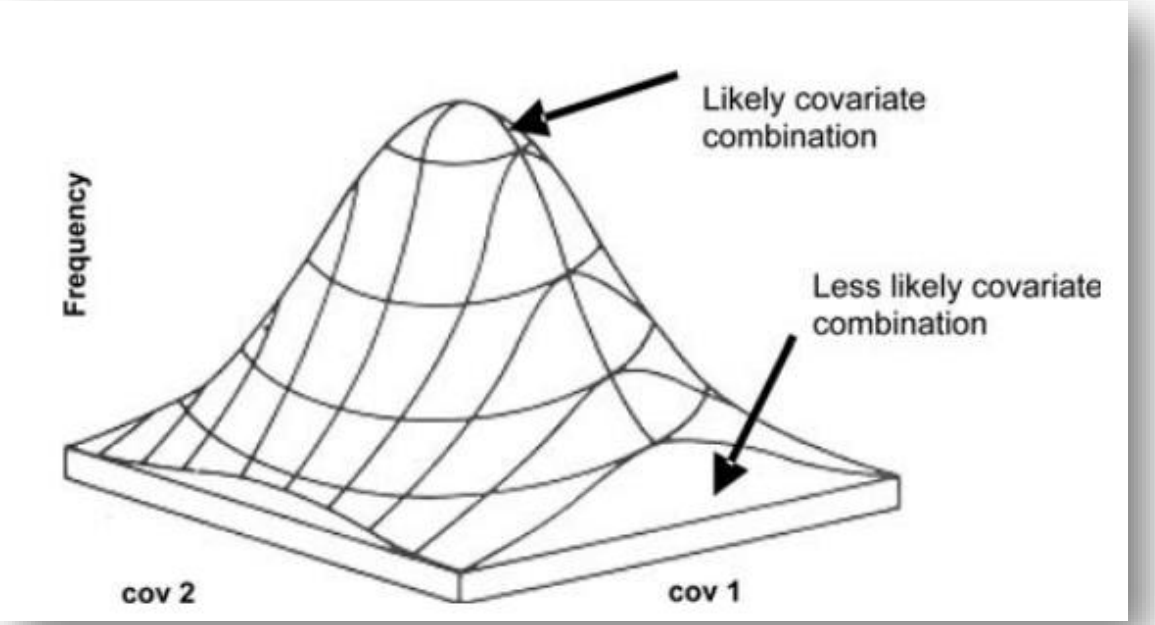
Marginal Distributions (Final simulation n=1)



Summary-Marginal Distributions

- Simulation ignoring correlations (i.e., joint distribution)
- Unrealistic simulation
  - Unrealistic combination of covariates
    - Low weight + high height
    - High age + high eGFR
- Easy to implement
- Not appropriate to use most of the time.

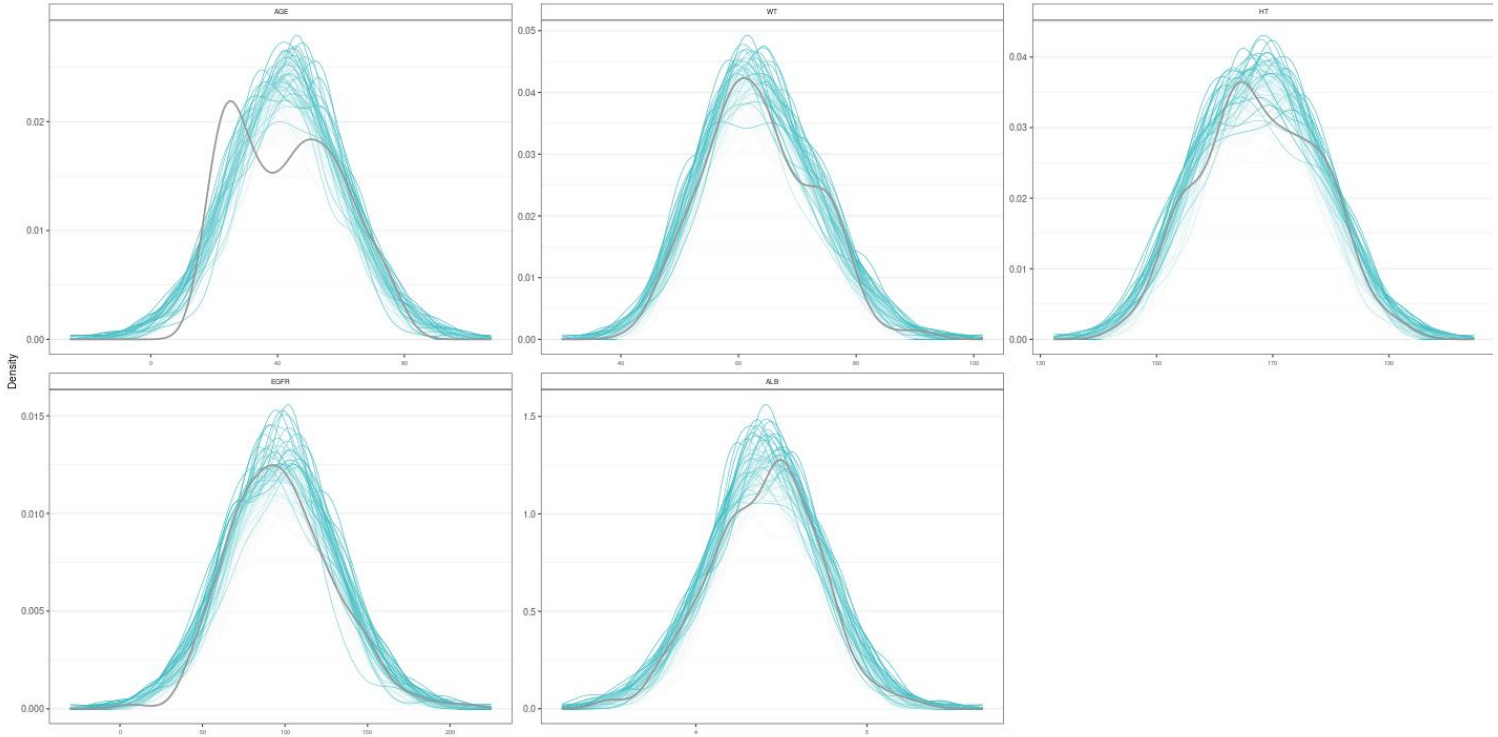
Multivariate Normal Distributions

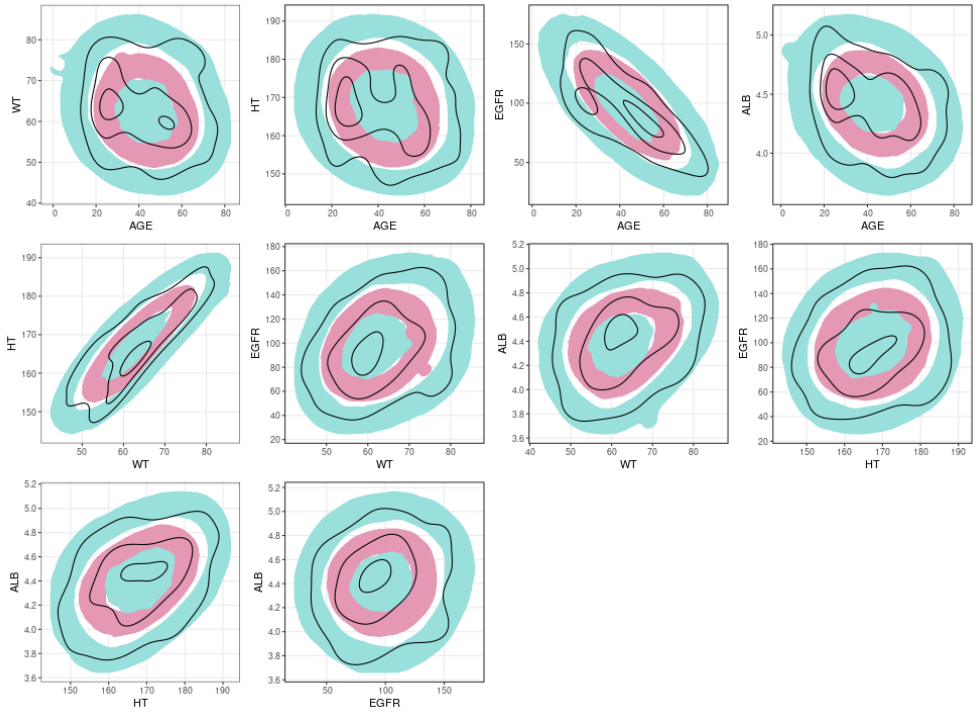


- Derive the means and the covariance matrix of the population dataset.
- Sample from the multivariate normal distribution defined by the means and the covariance matrix .

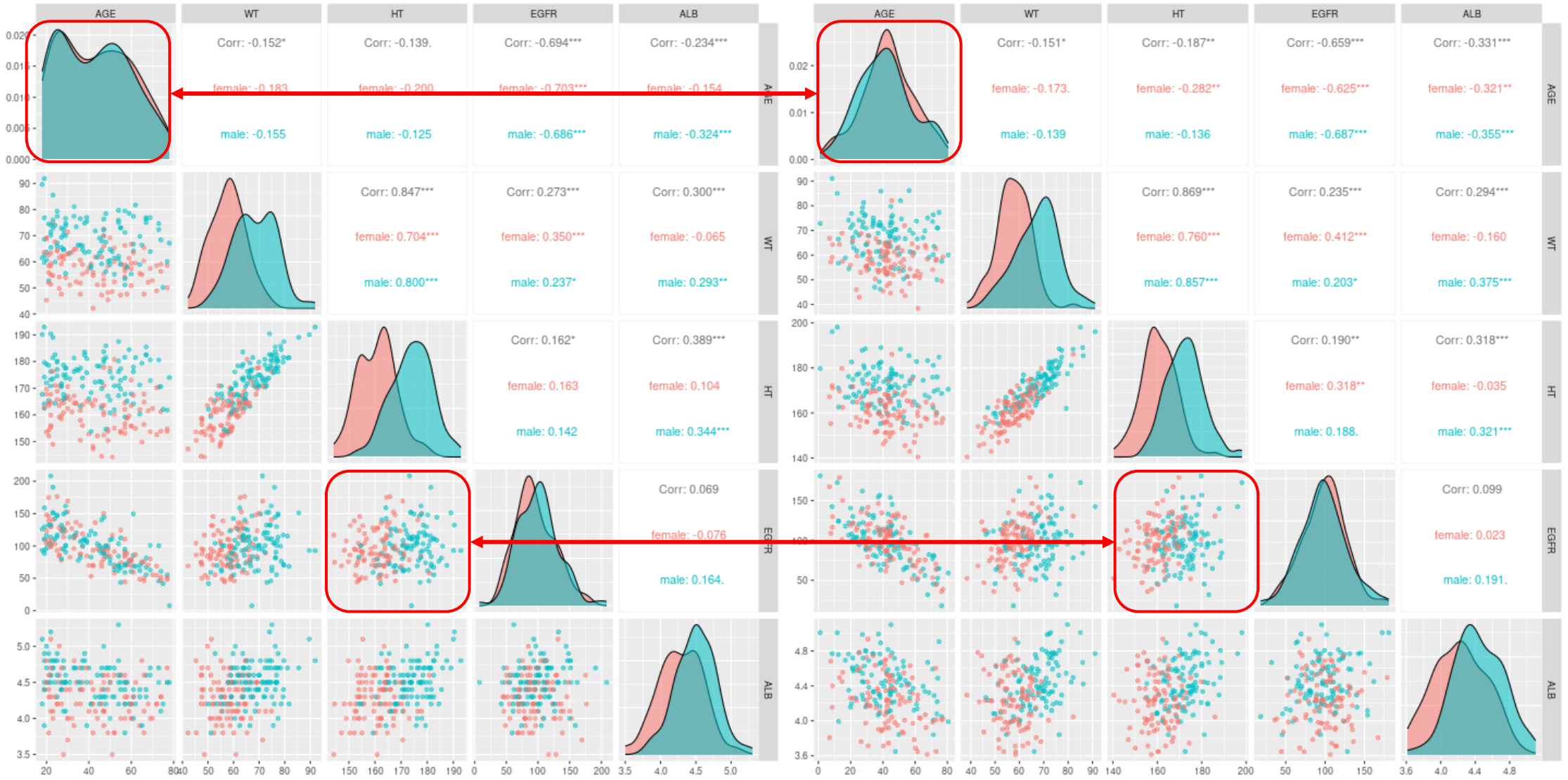
Tannenbaum et al. 2006. Journal of Pharmacokinetics and Pharmacodynamics

Multivariate Normal Distributions Simulation Method Evaluation (Simulation replicates: n=100)





Multivariate Normal Distributions (Final simulation n=1)



Summary-Multivariate Normal Distributions

- Simulation considering both marginal and joint distribution
- More realistic simulation
  - Realistic combination of covariates
- Strong distributional assumptions
  - Normal marginal distribution
  - Linear correlation

Copula-Based Virtual Patient Simulation

- Zwep L.B. “Virtual patient simulation: a copula approach.” *PAGE* (2022).
- Zwep, L. B., et al. “Virtual patient simulation using copula modeling.” *CPT* (2022).
- Guo, Y., et al. “Generation of realistic virtual adult populations using a model-based copula approach.” *JPKPD* (2024).
- Guo, Y., et al. “Copulas for Covariate Simulation in Pharmacometrics” *CPT:PSP* (2026)
  - R package (pmxcopula): https://github.com/vanhasseltlab/pmxcopula
  - Installation: `devtools::install_github("vanhasseltlab/pmxcopula", build_vignettes = TRUE, force = TRUE)`



TUTORIAL OPEN ACCESS

Copulas for Covariate Simulation in Pharmacometrics

Yuchen Guo | Tingjie Guo | J. G. Coen van Hasselt | Laura B. Zwep

Division of Systems Pharmacology and Pharmacy, Leiden Academic Center for Drug Research, Leiden University, Leiden, the Netherlands

Sklar's Theorem

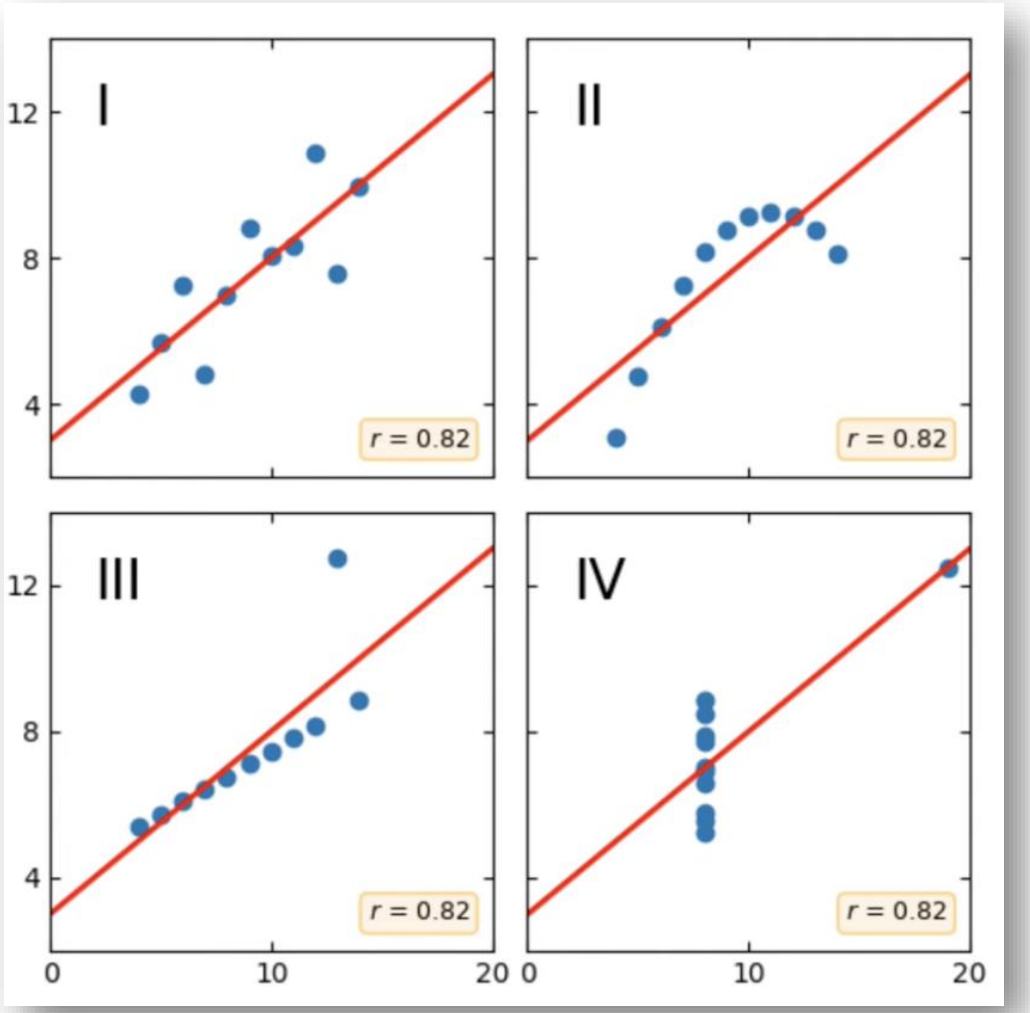
$$f(x,y) = C(U,V) \cdot f_X(X) \cdot f_Y(Y)$$

$$U = F_X(X) \quad U \sim U(0,1)$$
$$V = F_Y(Y) \quad V \sim U(0,1)$$
$$X = F_X^{-1}(U) \quad Y = F_Y^{-1}(V)$$

f (x, y)	Joint density function
Fx(X), FY (Y)	Marginal distribution
C (U, V)	Copula describing the dependence structure
$F_X^{-1}(X)$, $F_Y^{-1}(Y)$	Inverse marginal cumulative density function

Guo, Y., et al. *CPT:PSP* (2026)

Why Copulas? Correlation Is Not Enough



Same Pearson correlation ≠ Same joint distribution

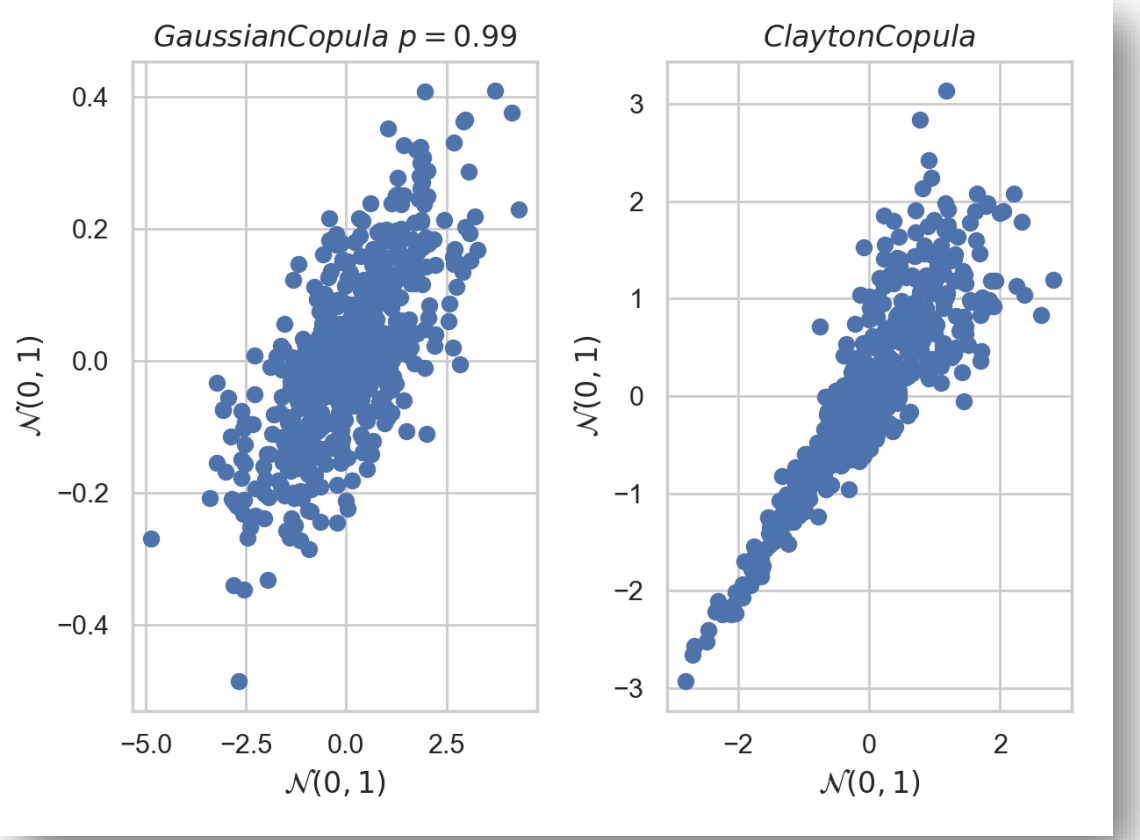
Datasets can have

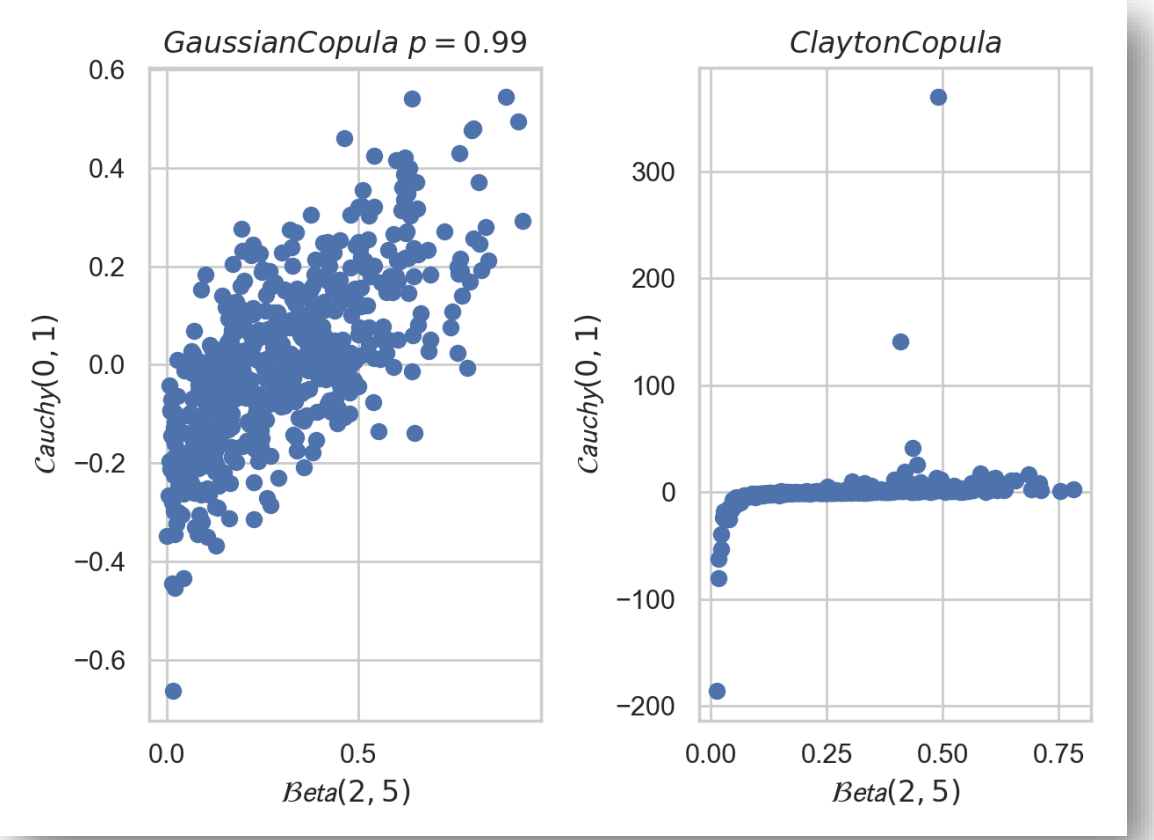
- The same correlation coefficient
- But differ substantially in their dependence patterns.

Matching a correlation matrix alone does not guarantee realistic combinations of patient covariates.

Kiran Karra. Copula Short Course. Youtube

Why copula? Dependence Beyond Correlation





Gaussian copula	Clayton copula
Symmetric dependence structure	Asymmetric dependence structure
No tail dependence	Lower tail dependence

Kiran Karra. Copula Short Course. Youtube

Copula-Based Simulation Workflow

$$f(x,y) = C(U_i, U_j) \times F_i(X_i) \times F_j(X_j)$$

1. Characterize observed covariates
Estimate marginal distribution of covariates (e.g., AGE, WT, EGFR, etc)

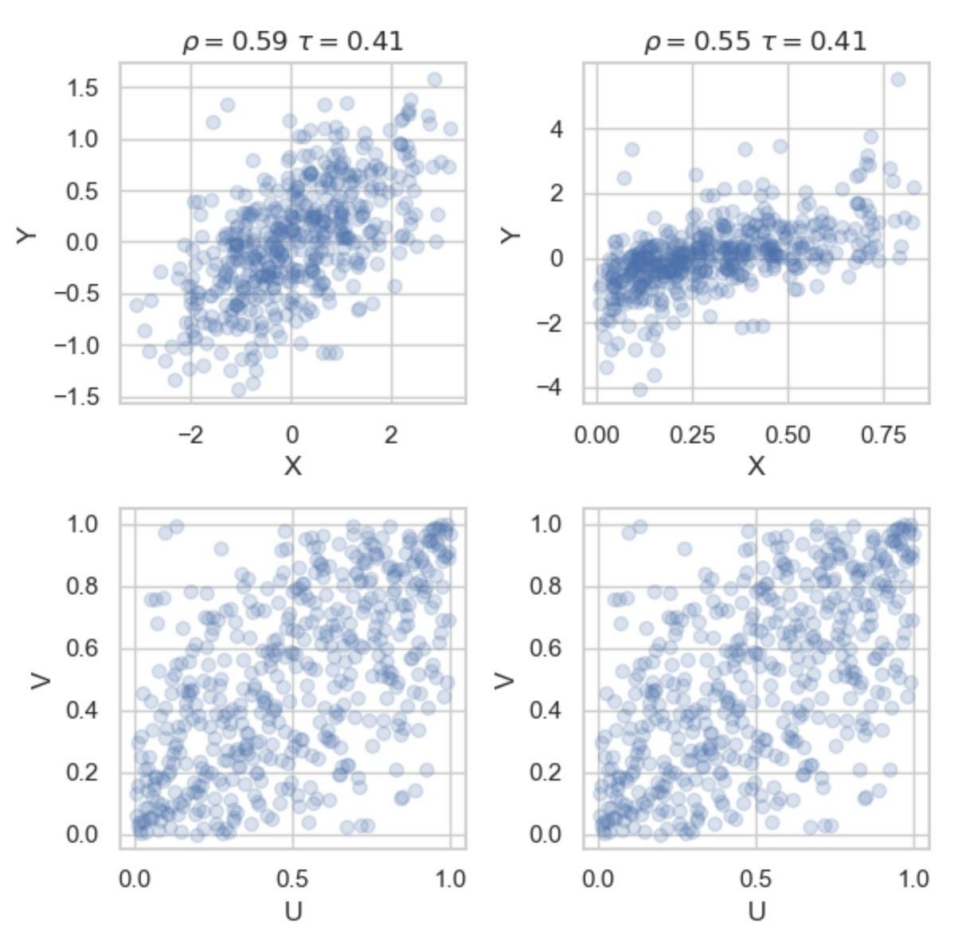
2. Transform covariates to uniform distributions
$U_j=F_j(X_j)$

3. Fit a copula model
Model dependence ($C(U_i, U_j)$) between uniformly distributed covariates
e.g., using vine copula guided by AIC

4. Simulate uniformly distributed covariates based on the fitted copula model
U_j^*

5. Transform back to clinical covariate values
$X_j^*=F_j^{-1}(U_j^*)$

Uniform Marginals Isolate the Dependence Structure



Why use uniform marginal distributions?

- Separate marginal distributions from dependence.
  - Model dependence regardless of the original covariate scales.
  - Preserve rank-based association.
- Generate new covariate combinations using different marginal distributions while retaining the fitted copula.

Original observations

x	y
0.6	0.8
0.2	0.4
1.2	0.5
0.1	0.2

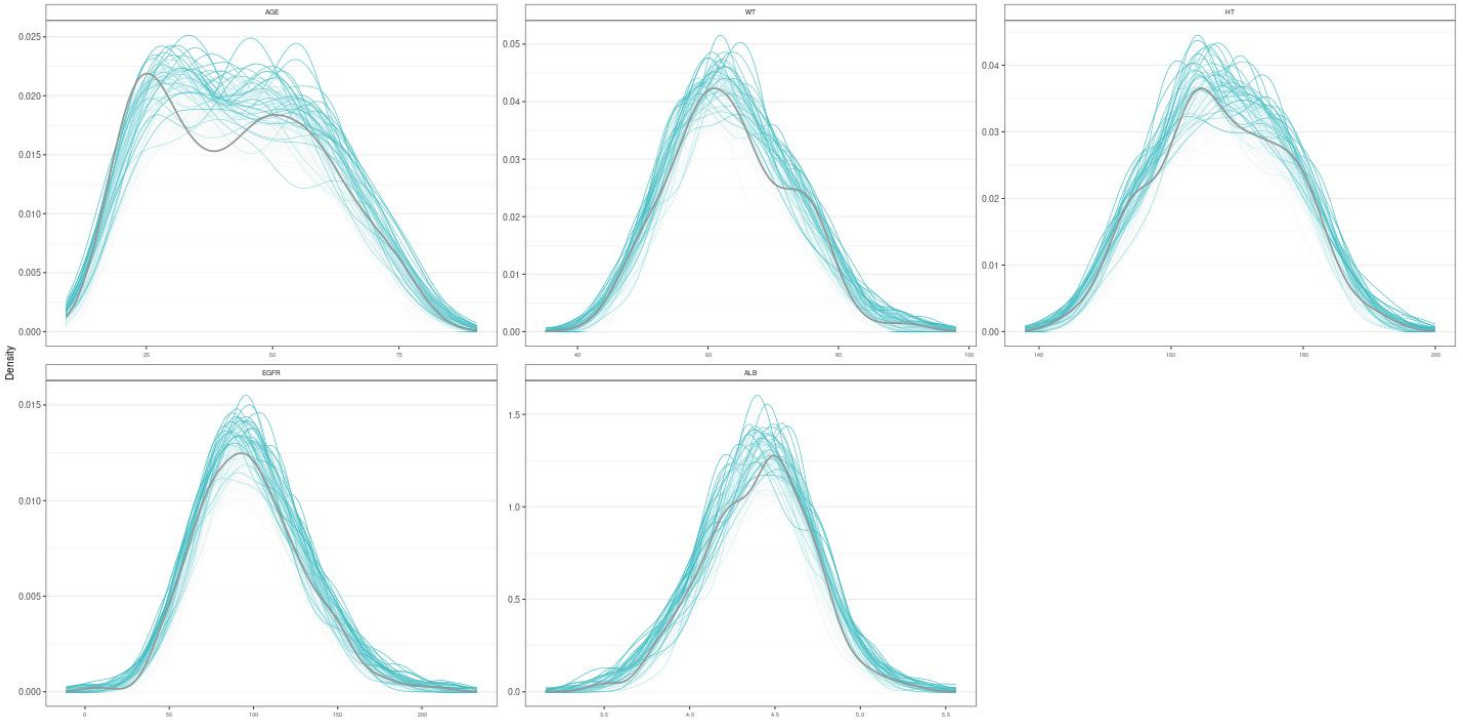
Rank observations

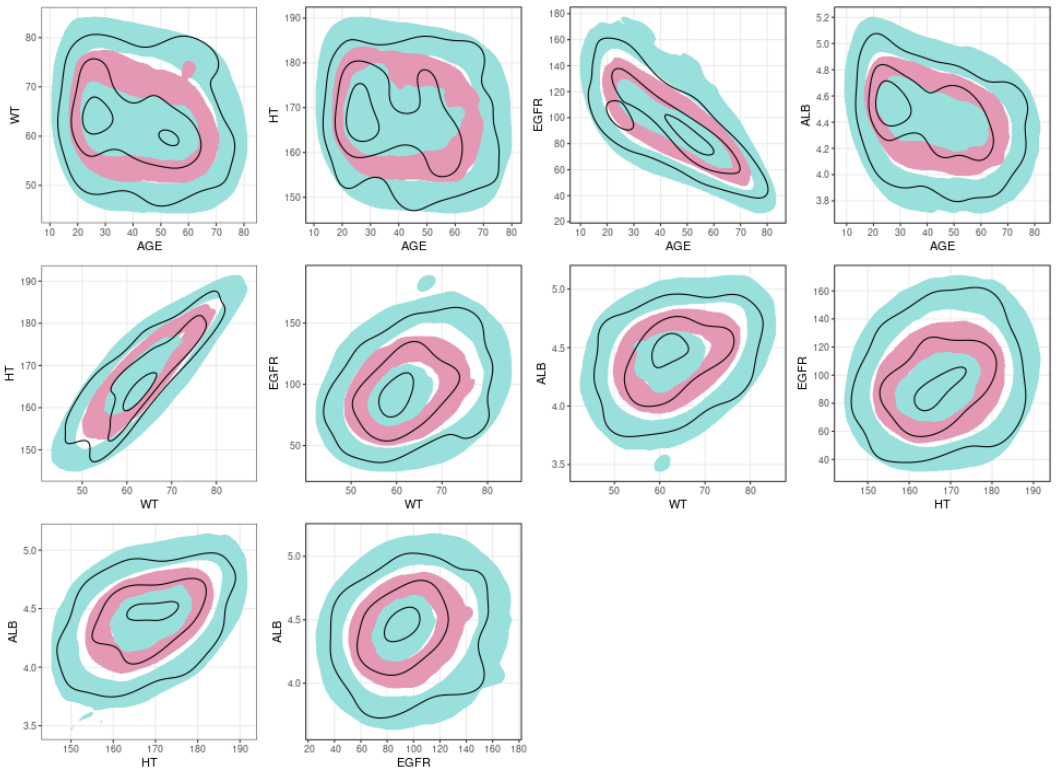
x	y
3	4
2	2
4	3
1	1

Kiran Karra. Copula Short Course. Youtube

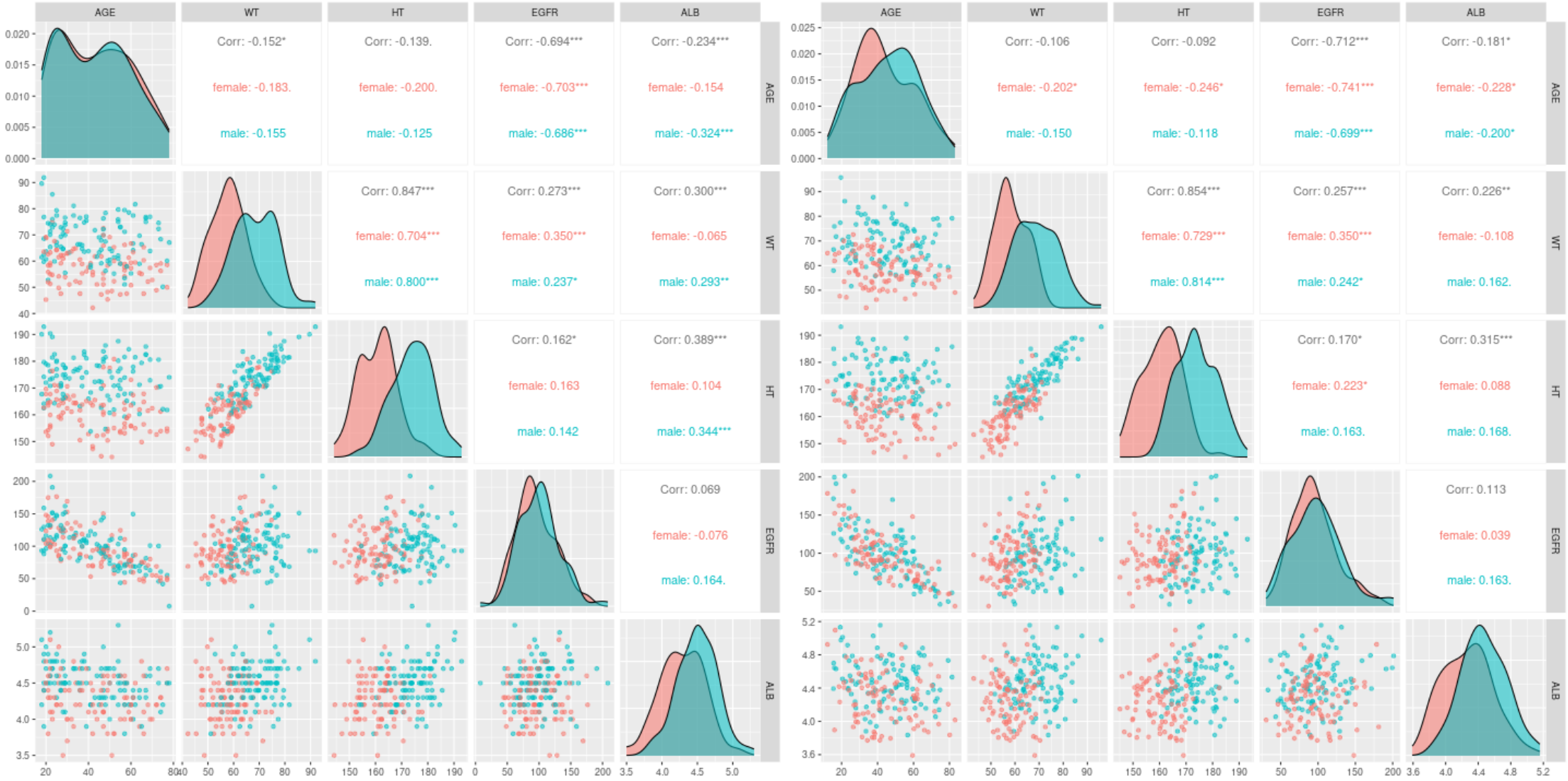
Key takeaway: Copulas allow us to model how patient covariates depend on one another, independently of how each covariate is distributed.

Copula Modeling Simulation Method Evaluation (Simulation replicates: n=100)





Copula Modeling (Final simulation n=1)



Copula resources

- Laura B. Zwep, Tingjie Guo, Jacqueline J. Meulman, J.G. Coen van Hasselt. Copula modeling for realistic virtual patient covariate simulation. PAGE 2022
- Zwep, L.B., Guo, T., Nagler, T., Knibbe, C.A.J., Meulman, J.J. and van Hasselt, J.G.C. (2024), Virtual Patient Simulation Using Copula Modeling. Clin Pharmacol Ther, 115: 795-804. https://doi.org/10.1002/cpt.3099
  - Github: https://github.com/vanhasseltlab/copula_vps
- Guo, Y., Guo, T., Knibbe, C.A.J. et al. Generation of realistic virtual adult populations using a model-based copula approach. J Pharmacokinet Pharmacodyn (2024). https://doi.org/10.1007/s10928-024-09929-4
  - Github: https://github.com/vanhasseltlab/NHANES_copula
  - ShinyAPP: https://cocosim.lacdr.leidenuniv.nl/
- Kiran Karra. Copula Short Course. Youtube

Homework

- Deterministic, stochastic, and virtual population simulations to support exposure-response analysis, exposure matching and dose recommendations.
  - Details: `./wk11/homework/homework-wk11.pdf`
  - All required documents are in `./wk11/homework/` directory
- Make a copy of `./wk11/ homework-wk11.qmd` with appropriate rename. Perform all analysis based on this qmd file. Knit into a pdf file for submission.
- Deadline: **Dec. 11, 2026** before class (3-week)
- This assignment will account for **20%** of the final grade.

The End